

RAHUL S TAWARAKHED

Final-Year B.E. (Artificial Intelligence & Machine Learning) | Python Developer | Software Engineering

The Oxford College of Engineering, Bengaluru

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PROFESSIONAL SUMMARY

Final-year B.E. Artificial Intelligence & Machine Learning student and Python Developer with hands-on experience building machine learning applications, web platforms, and AI-powered automation systems. Proficient in Python, Flask, Scikit-learn, JavaScript, databases, and Git/GitHub, with experience developing and deploying real-world software solutions. Strong problem-solving, communication, teamwork, and leadership skills with a keen interest in software engineering and intelligent application development.

EDUCATION

The Oxford College of Engineering, Bengaluru | 2023 - Present

B.E. - Artificial Intelligence and Machine Learning | CGPA: 8.604 / 10 (up to 6th Semester)

SMT. Pusha Mahalingappa PU College | 2021 - 2023

Class XII (PUC) – PCMB

Adarsha Vidyalaya (RMSA), Atanoor, Kalaburagi | 2019 - 2021

Class X | 94.2%

SKILLS

Programming Languages: Python, Java, C, JavaScript

Backend & Web: Flask, FastAPI, ReactJS, HTML, CSS, REST APIs

Databases: MySQL, PostgreSQL, MongoDB, SQLite, SQL

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Machine Learning: Scikit-learn, TensorFlow, Keras, EfficientNetB0

Libraries: Pandas, NumPy, OpenCV, Requests

Developer Tools: Git, GitHub, VS Code, Render, Vercel

Additional Technologies: Tkinter, SpeechRecognition, PyAudio, psutil, Threading

Soft Skills: Problem-solving, Leadership, Communication, Teamwork

EXPERIENCE

Web Development Intern - Skill Craft Technology | August 2025 - September 2025

- Built and deployed 4 interactive web applications using HTML, CSS, and JavaScript, implementing responsive layouts, DOM manipulation, event-driven functionality, and user-focused interfaces.
- Applied responsive web-development principles to build functional and user-friendly browser-based applications.
- Strengthened practical experience in frontend development, debugging, project implementation, and software development

Smart India Hackathon (SIH) - Participant | 2025

- Collaborated on innovative technology solutions under strict development timelines in a competitive hackathon environment.
- Gained hands-on experience in problem-solving, teamwork, rapid prototyping, and solution development.

PROJECTS

AI Desktop Assistant – Voice-Controlled System Automation [Github-Repo]

- Developed NeuroDesk, an adaptive AI desktop assistant supporting text and voice-based system automation, including application launching/closing, indexed file search, system utilities, and resource monitoring.
- Implemented a hybrid intent-routing architecture combining deterministic rules with TF-IDF + Logistic Regression, achieving 90% intent-classification accuracy across 15 command categories.
- Integrated real-time CPU, RAM, disk, battery, disk-I/O, and process-memory monitoring with adaptive Balanced, Performance, and Power-Save modes.

Tech Stack: Python, Tkinter, Scikit-learn, TF-IDF, Logistic Regression, SpeechRecognition, SoundDevice, SQLite, psutil, Threading, Windows Automation, Git/GitHub

GreenGrow AI - Fertilizer Recommendation System [GitHub-Repo] [Live-Demo]

- Developed an ML-based fertilizer recommendation application using soil nutrient values such as Nitrogen, Phosphorus, Potassium, and environmental parameters.
- Trained and deployed a machine-learning model achieving 88% prediction accuracy across 6 fertilizer categories.
- Integrated Flask for backend processing and developed a user-friendly interface for real-time fertilizer recommendations.

Tech Stack: Python, Streamlit, Flask, HTML, CSS, Scikit-learn, Pandas, NumPy, SQLite, Git/GitHub

Home Made Products - E-Commerce Web Application [GitHub-Repo] [Live-Demo]

- Developed a web-based e-commerce platform for browsing and managing homemade products through a clean and intuitive user interface.
- Implemented product listing, product-detail views, and streamlined navigation to improve the customer browsing experience.
- Designed an extensible application architecture supporting product management and order-processing functionality.

Tech Stack: Python, Flask, SQLite, HTML, CSS, JavaScript, Render

CERTIFICATES & ACHIEVEMENTS

- Python for Data Science - NPTEL Course Completion Certificate (2025)
- Internship Completion Certificate - Skill Craft Technology (2025)
- Smart India Hackathon (SIH) - Participant (2025)
- SAP HackFest - Participant (2025)
- Sentinel Hackathon - Participant (2025)